

Interaction picture

The interaction picture is a hybrid of the Schrödinger and Heisenberg pictures. In it, states evolve with the interaction Hamiltonian and operators evolve with the free Hamiltonian. As we will see, this allows us to use time-dependent perturbation theory in a more convenient fashion. The free Hamiltonian will trivially evolve operators, while the interaction Hamiltonian will evolve states. In this language, the state evolution will be treated perturbatively, while the operator evolution is assumed to be known exactly.

We start with a Hamiltonian split into a free part $\hat{H}_0$ and an interaction part $V(t)$, where $\hat{H}_0$ is time-independent:

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t). \quad (1)$$

The interaction picture states are defined by

$$|\psi_I(t)\rangle = e^{\frac{i}{\hbar}\hat{H}_0 t} |\psi_S(t)\rangle. \quad (2)$$

We can think of this state as “unwinding” the free evolution from the Schrödinger picture state, leaving only the interaction evolution to be treated. Note that because the hamiltonian is time-independent, the free evolution operator is just an exponential (no integrals or time-ordering). In addition, note that for $t = 0$, the interaction picture state coincides with the Schrödinger picture state:

$$|\psi_I(0)\rangle = |\psi_S(0)\rangle. \quad (3)$$

The interaction picture operators are defined by the opposite transformation:

$$\hat{O}_I(t) = e^{\frac{i}{\hbar}\hat{H}_0 t} \hat{O}_S e^{-\frac{i}{\hbar}\hat{H}_0 t}. \quad (4)$$

Note this is just the Heisenberg picture evolution of the operator with respect to the free Hamiltonian $\hat{H}_0$.

While before, in the two previous pictures only one of the objects evolved in time, now both states and operators evolve in time (though via different Hamiltonians). In particular, the interaction picture Hamiltonian is

$$\hat{H}_I(t) = e^{\frac{i}{\hbar}\hat{H}_0 t} \hat{H}(t) e^{-\frac{i}{\hbar}\hat{H}_0 t} = \hat{H}_0 + e^{\frac{i}{\hbar}\hat{H}_0 t} \hat{V}(t) e^{-\frac{i}{\hbar}\hat{H}_0 t} = \hat{H}_0 + \hat{V}_I(t), \quad (5)$$

where we used the fact that H_0 commutes with itself and have defined the interaction picture “interaction” $\hat{V}_I(t)$:

$$\hat{V}_I(t) = e^{\frac{i}{\hbar}\hat{H}_0 t} \hat{V}(t) e^{-\frac{i}{\hbar}\hat{H}_0 t}. \quad (6)$$

?

Prove that the Schrodinger equation in the interaction picture is just

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = \hat{V}_I(t) |\psi_I(t)\rangle. \quad (7)$$

Prove also that the Heisenberg equation of motion for an operator $\hat{O}_I(t)$ is

$$i\hbar \frac{d}{dt} \hat{O}_I(t) = [\hat{O}_I(t), \hat{H}_0]. \quad (8)$$

which differ from the Schrodinger and Heisenberg equations of motion by the replacement $\hat{H} \rightarrow \hat{V}_I$ and $\hat{H} \rightarrow \hat{H}_0$, respectively.

Eq. (7) is a much simpler equation than the original Schrodinger equation, as it only involves the interaction Hamiltonian, which we will treat perturbatively. In some sense, we have hidden

the free evolution into the operators and can now focus on how the interaction evolves the state. If you stare at this equation long enough, you will see that it is equivalent to the original Schrodinger equation. If we find a given solution $\psi_I(t)$, then all we have to do is apply the free evolution to get the Schrodinger picture state, namely, invert Eq. (2), $|\psi_S(t)\rangle = e^{-\frac{i}{\hbar}\hat{H}_0 t} |\psi_I(t)\rangle$.

Let us investigate the time dependence of some arbitrary state

$$|\psi_I(t)\rangle = \sum_k c_k(t) |k\rangle, \quad (9)$$

where $|k\rangle$ are the eigenstates of the free Hamiltonian $\hat{H}_0$. Plugging this into the Schrodinger equation in the interaction picture, we get

$$i\hbar \sum_k \dot{c}_k(t) |k\rangle = \sum_k c_k(t) \hat{V}_I(t) |k\rangle. \quad (10)$$

Multiplying by $\langle n|$, we get

$$i\hbar \dot{c}_n(t) = \sum_k c_k(t) \langle n| \hat{V}_I(t) |k\rangle = \sum_k c_k(t) e^{i w_{nk} t} V_{nk}, \quad (11)$$

where we used our definitions $w_{nk} = (E_n - E_k)/\hbar$ and $V_{nk} = \langle n| \hat{V} |k\rangle$. (Caution here! V_{nk} is not the matrix element of $\hat{V}_I$ but of $\hat{V}$.) As before, this is a set of coupled first-order differential equations for the coefficients $c_n(t)$, which we can solve perturbatively. Recall that before we had arrived at this equation with the equivalent ansatz $|\psi_S(t)\rangle = \sum_k c_k(t) e^{-\frac{i}{\hbar} E_k t} |k\rangle$.

By the way, if we were to go back and carry out our perturbative expansion $c_n(t) = c_n^{(0)}(t) + c_n^{(1)}(t) + \dots$, we could now write the order-by-order equations in a somewhat simpler way. Let us recall that procedure here. We start with an initial state $|\psi(0)\rangle = |i\rangle$, which means that $c_n(0) = \delta_{ni}$. The zeroth order equation is just

$$\dot{c}_n^{(0)}(t) = 0, \quad (12)$$

which means that without the perturbation, the state remains in the initial state $|i\rangle$, as expected. The first order equation gives

$$\dot{c}_n^{(1)}(t) = -\frac{i}{\hbar} \sum_k c_k^{(0)}(t) \langle n| \hat{V}_I(t) |k\rangle \quad (13)$$

$$\Rightarrow c_n^{(1)}(t) = -\frac{i}{\hbar} \sum_k \int_0^t dt' c_k^{(0)}(t') \langle n| \hat{V}_I(t') |k\rangle = -\frac{i}{\hbar} \int_0^t dt' \langle n| \hat{V}_I(t') |i\rangle. \quad (14)$$

Previously, we worked with a potential that was zero for $t < 0$, and then turned on as a constant for $t \geq 0$, to show that this solution gave picked transitions between the initial state and other states with the same energy. The same can be done here by using Eq. (6). The second order equation gives

$$\dot{c}_n^{(2)}(t) = -\frac{i}{\hbar} \sum_k c_k^{(1)}(t) \langle n| \hat{V}_I(t) |k\rangle \quad (15)$$

$$\Rightarrow c_n^{(2)}(t) = -\frac{i}{\hbar} \sum_k \int_0^t dt' c_k^{(1)}(t') \langle n| \hat{V}_I(t') |k\rangle \quad (16)$$

$$= \left(-\frac{i}{\hbar}\right)^2 \sum_k \int_0^t dt' \int_0^{t'} dt'' \langle n| \hat{V}_I(t') |k\rangle \langle k| \hat{V}_I(t'') |i\rangle \quad (17)$$

$$= \frac{1}{2} \left(-\frac{i}{\hbar}\right)^2 \int_0^t dt' \int_0^t dt'' \langle n| \mathcal{T} \{ \hat{V}_I(t') \hat{V}_I(t'') \} |i\rangle, \quad (18)$$

where we used the time-ordering operator $\mathcal{T}$ to write the last line in a more compact way. You might see where this is going, but before we get to the punchline, let us discuss the formalism a bit more.

Time-Evolution Operator

We also ought to define an interaction picture time-evolution operator $\hat{U}_I(t, t_0)$. We hope it respects its own Schrodinger equation (in the interaction picture), namely,

$$i\hbar \frac{\partial}{\partial t} \hat{U}_I(t, t_0) = \hat{V}_I(t) \hat{U}_I(t, t_0). \quad (19)$$

In a small step dt , we have

$$\hat{U}_I(t + dt, t) = \mathbb{1} - \frac{i}{\hbar} \hat{V}_I(t) dt, \quad (20)$$

Just as we did back in the time-evolution operator section, we can integrate this equation to get an integral equation for $\hat{U}_I$. Again, we may not necessarily have that $V(t)$ commutes with itself at different times, so we have to be careful about the order of the operators when integrating. Carrying out exactly the same steps as before (see Appendix B of Time-Evolution notes), namely, writing

$$\hat{U}_I(t_N, t_0) = \hat{U}_I(t_N, t_{N-1}) \hat{U}_I(t_{N-1}, t_{N-2}) \cdots \hat{U}_I(t_1, t_0), \quad (21)$$

and taking the limit $N \rightarrow \infty$ ($t_\infty \rightarrow t$), we get the Dyson series for the interaction picture time-evolution operator:

$$\hat{U}_I(t, t_0) = \mathbb{1} + \sum_{n=1}^{\infty} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_n \int_{t_0}^{t_n} dt_{n-1} \cdots \int_{t_0}^{t_2} dt_1 \hat{V}_I(t_n) \hat{V}_I(t_{n-1}) \cdots \hat{V}_I(t_1). \quad (22)$$

where we pulled out the identity term explicitly to make the $n = 0$ term more clear. Using the definition of the time-ordering operator $\mathcal{T}$:

$$\boxed{\hat{U}_I(t, t_0) = \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_0}^t dt' \hat{V}_I(t') \right]}, \quad (23)$$

where $\mathcal{T}$ is the time-ordering operator. This is a Dyson series and is equivalent to our previous time-evolution operator with the “unwinding” of the free evolution.



The time evolution operator does not transform the same way as other operators under the interaction picture transformation. Show that

$$\hat{U}_I(t, t_0) = e^{\frac{i}{\hbar} \hat{H}_0 t} \hat{U}(t, t_0) e^{-\frac{i}{\hbar} \hat{H}_0 t_0}, \quad (24)$$

where $\hat{U}(t, t_0)$ is the Schrodinger picture time-evolution operator. Use the definition of the interaction picture and the Schrodinger picture states.



Note that despite this funny transformation, the interaction picture time-evolution operator still satisfies the usual composition property $\hat{U}_I(t_2, t_0) = \hat{U}_I(t_2, t_1) \hat{U}_I(t_1, t_0)$, as well as the initial condition $\hat{U}_I(t_0, t_0) = \mathbb{1}$. Another interesting observation is that if we take two energy eigenstates $|i\rangle$ and $|f\rangle$, the transition between them is the same whether we evolve $|i\rangle$ with the Schrodinger picture or in the interaction picture. To see that, note that

$$\langle f | \hat{U}_I(t, t_0) | i \rangle = e^{\frac{i}{\hbar} E_f t} \langle f | \hat{U}(t, t_0) | i \rangle e^{-\frac{i}{\hbar} E_i t_0}, \quad (25)$$

so the probability $|\langle f | \hat{U}_I(t, t_0) | i \rangle|^2$ is the same as $|\langle f | \hat{U}(t, t_0) | i \rangle|^2$.

Propagators

Recall the definition of the propagator (in position space):

$$K(x, t; x_0, t_0) = \langle x | \hat{U}(t, t_0) | x_0 \rangle, \quad (26)$$

where we interpret it as the amplitude for an infinitely-well-localized particle to go from x_0 at time t_0 to x at time t (for a discussion of the propagator as the Green's function of the Schrodinger equation, see the Green's functions notes). It had a couple of nice properties, including the composition property

$$K(x, t; x_0, t_0) = \int dx' K(x, t; x', t') K(x', t'; x_0, t_0). \quad (27)$$

We interpret this as the following: the amplitude for a particle to go from x_0 at time t_0 to x at time t is the sum over all possible intermediate positions x' at an intermediate time t' of the amplitude to go from x_0 to x' and then from x' to x . The propagator is subject to the initial condition

$$K(x, t_0; x_0, t_0) = \delta(x - x_0). \quad (28)$$

Note also that the conjugate of the propagator is the amplitude for the reverse process, namely, to go from x at time t to x_0 at time t_0 :

$$K^*(x, t; x_0, t_0) = \langle x_0 | \hat{U}^\dagger(t, t_0) | x \rangle = \langle x_0 | \hat{U}(t_0, t) | x \rangle = K(x_0, t_0; x, t). \quad (29)$$

By extension, we can also define the free and the interaction picture propagators, respectively,

$$K_0(x, t; x_0, t_0) = \langle x | e^{-\frac{i}{\hbar} \hat{H}_0(t-t_0)} | x_0 \rangle, \quad \text{and} \quad K_I(x, t; x_0, t_0) = \langle x | \hat{U}_I(t, t_0) | x_0 \rangle. \quad (30)$$

?

Using the transformation of $\hat{U}_I$ in terms of $\hat{U}$, we prove the following relation between propagators:

$$\begin{aligned} K_I(x, t; x_0, t_0) &= \int dx' \int dx'_0 K_0^*(x', t; x, 0) K(x', t; x'_0, t_0) K_0(x'_0, t_0; x_0, 0) \\ &= \int dx' \int dx'_0 K_0(x, 0; x', t) K(x', t; x'_0, t_0) K_0(x'_0, t_0; x_0, 0). \end{aligned} \quad (31)$$

Pay close attention to the subscripts here. Show that the interaction picture propagator satisfies all the same properties as the full propagator, including the composition property and the initial condition. For example,

$$K_I(x, t; x_0, t_0) \xrightarrow{t \rightarrow t_0} K_0(x, 0; x_0, 0) = \delta(x - x_0), \quad (32)$$

so the interaction picture propagator reduces to the free propagator in the limit where the initial and final times coincide.

Note that when we turn off the interaction ($\hat{V} = 0$), K reduces to K_0 . However, K_I does not reduce to K_0 when we turn off the interaction, but rather to the identity operator in position space, which is just a delta function:

$$K_I(x, t; x_0, t_0) \xrightarrow{\hat{V} \rightarrow 0} \langle x | \mathbb{1} | x_0 \rangle = \delta(x - x_0). \quad (33)$$

You can also show this with the composition property using Eq. (31). So, in the absence of interactions, K_I does not propagate the state at all, directly analogous to how $U_I \xrightarrow{\hat{V} \rightarrow 0} \mathbb{1}$.



Conversely, prove that the full propagator can also be written in terms of the interaction-picture propagator as

$$\begin{aligned} K(x, t; x_0, t_0) &= \int dx' \int dx'_0 K_0(x, t; x', 0) K_I(x', t; x'_0, t_0) K_0^*(x_0, t_0; x'_0, 0) \\ &= \int dx' \int dx'_0 K_0(x, t; x', 0) K_I(x', t; x'_0, t_0) K_0(x'_0, 0; x_0, t_0). \end{aligned} \quad (34)$$

Dyson series for the propagator

Because the propagator is just a transition amplitude for the time evolution operator, we can also write the Dyson series for the propagator by taking the matrix element of the Dyson series for $\hat{U}_I$. Using Eq. (22), we get

$$K_I(x, t; x_0, t_0) = \sum_{n=0}^{\infty} \left(-\frac{i}{\hbar} \right)^n \int_{t_0}^t dt_n \int_{t_0}^{t_n} dt_{n-1} \cdots \int_{t_0}^{t_2} dt_1 \langle x | \hat{V}_I(t_n) \hat{V}_I(t_{n-1}) \cdots \hat{V}_I(t_1) | x_0 \rangle, \quad (35)$$

This is nice and exactly what we expected out of the interaction picture, but not terribly useful since what we care about usually is the full evolution of the state, obtained with the Schrodinger picture propagator K .

To get the Dyson series for K , we will go back to the series for $\hat{U}_I$, namely Eq. (22), and insert it inside the definition

$$K(x, t; x_0, t_0) = \langle x | e^{-\frac{i}{\hbar} \hat{H}_0 t} \hat{U}_I(t, t_0) e^{\frac{i}{\hbar} \hat{H}_0 t_0} | x_0 \rangle. \quad (36)$$

Expanding each $\hat{V}_I(t_k)$ as in Eq. (6), we get that the adjacent potentials inside the Dyson series are sandwiched between free evolution operators in the form of

$$e^{-\frac{i}{\hbar} \hat{H}_0(t-t_n)} \hat{V}(t_n) e^{-\frac{i}{\hbar} \hat{H}_0(t_n-t_{n-1})} \hat{V}(t_{n-1}) \cdots \hat{V}(t_1) e^{-\frac{i}{\hbar} \hat{H}_0(t_1-t_0)}, \quad (37)$$

where the outermost exponentials come from Eq. (36). The nested time structure is inherited from the integration limits of Eq. (22).

We go further: to make the evolution in space more explicit, we insert n position completeness relations, one between each free-evolution operator and each potential. In principle, we would need $2n$ such relations, one for each side of V for n terms. However, for a *local* potential (one that is diagonal in position space), we have

$$\langle x' | \hat{V}(t) | x'' \rangle = V(x'', t) \delta(x' - x''), \quad (38)$$

and so we get to trivially integrate n out of the $2n$ position variables. The n -th order term then becomes

$$\begin{aligned} K^{(n)}(x, t; x_0, t_0) &= \left(-\frac{i}{\hbar} \right)^n \int_{t_0}^t dt_n \int_{t_0}^{t_n} dt_{n-1} \cdots \int_{t_0}^{t_2} dt_1 \int dx_n \cdots \int dx_1 \\ &\quad \times K_0(x, t; x_n, t_n) V(x_n, t_n) K_0(x_n, t_n; x_{n-1}, t_{n-1}) V(x_{n-1}, t_{n-1}) \\ &\quad \times \cdots V(x_1, t_1) K_0(x_1, t_1; x_0, t_0), \end{aligned} \quad (39)$$

where $K_0(x_b, t_b; x_a, t_a) = \langle x_b | e^{-\frac{i}{\hbar} \hat{H}_0(t_b-t_a)} | x_a \rangle$ is the free propagator. Note the physical picture encoded in this expression: the particle propagates freely from (x_0, t_0) to some intermediate point (x_1, t_1) , experiences the potential $V(x_1, t_1)$, then propagates freely to (x_2, t_2) , experiences the potential $V(x_2, t_2)$, and so on, until it finally propagates freely to (x, t) . Each such “encounter” with the potential is called a scattering event, and the n -th order term represents all processes in which the particle undergoes exactly n scatterings.

The full Dyson series for the propagator is then

$$K(x, t; x_0, t_0) = K_0(x, t; x_0, t_0) + \quad (40)$$

$$\sum_{n=1}^{\infty} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_n \cdots \int_{t_0}^{t_2} dt_1 \int dx_n \cdots dx_1 \prod_{k=0}^n K_0(x_{k+1}, t_{k+1}; x_k, t_k) \prod_{j=1}^n V(x_j, t_j), \quad (41)$$

where we adopt the notational convention $x_{n+1} \equiv x$, $t_{n+1} \equiv t$. This is also known as the *Born* series. You may have heard of the Born approximation, which is quite literally the approximation of truncating this series at the first order, i.e., $K \approx K_0 + K^{(1)}$.

Let us write out the first few orders explicitly. The zeroth order is simply the free propagator:

$$K^{(0)}(x, t; x_0, t_0) = K_0(x, t; x_0, t_0). \quad (42)$$

The first order correction introduces a single scattering event:

$$K^{(1)}(x, t; x_0, t_0) = -\frac{i}{\hbar} \int_{t_0}^t dt_1 \int dx_1 K_0(x, t; x_1, t_1) V(x_1, t_1) K_0(x_1, t_1; x_0, t_0). \quad (43)$$

The second order introduces two scattering events:

$$\begin{aligned} K^{(2)}(x, t; x_0, t_0) &= \left(-\frac{i}{\hbar}\right)^2 \int_{t_0}^t dt_2 \int_{t_0}^{t_2} dt_1 \int dx_2 \int dx_1 \\ &\quad \times K_0(x, t; x_2, t_2) V(x_2, t_2) K_0(x_2, t_2; x_1, t_1) V(x_1, t_1) K_0(x_1, t_1; x_0, t_0). \end{aligned} \quad (44)$$

Using the time-ordering operator, each term in the series can be written in the more compact form

$$\begin{aligned} K^{(n)}(x, t; x_0, t_0) &= \frac{1}{n!} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_n \cdots \int_{t_0}^{t_2} dt_1 \int dx_n \cdots dx_1 \\ &\quad \langle x | \mathcal{T} \left\{ e^{-\frac{i}{\hbar} \hat{H}_0(t-t_n)} \hat{V}(t_n) \cdots \hat{V}(t_1) e^{-\frac{i}{\hbar} \hat{H}_0(t_1-t_0)} \right\} | x_0 \rangle. \end{aligned} \quad (45)$$

Recall from our Green's functions notes that it was the *retarded* propagator that served as the Green's function of the Schrodinger equation, so technically the Dyson series we have derived here is for the retarded propagator, which is the one that satisfies the initial condition $K(x, t_0^+; x_0, t_0) = \delta(x - x_0)$.

?

Verify explicitly that the first-order Born term $K^{(1)}$ satisfies an integral equation of the form

$$K(x, t; x_0, t_0) = K_0(x, t; x_0, t_0) - \frac{i}{\hbar} \int_{t_0}^t dt' \int dx' K_0(x, t; x', t') V(x', t') K(x', t'; x_0, t_0), \quad (46)$$

by iterating the right-hand side perturbatively (i.e., replacing K on the right by $K_0 + K^{(1)} + \cdots$ and collecting terms order by order). This is the Lippmann–Schwinger equation for the propagator; the full Born series Eq. (40) is its Neumann-series solution.

Non-local potentials

If the potential is *not* local in position space, e.g., a momentum-dependent interaction with

$$\langle x' | \hat{V}(t) | x'' \rangle \neq V(x'', t) \delta(x' - x''), \quad (47)$$

then each vertex contributes *two* position integrations, and the n -th order term reads

$$\begin{aligned}
K^{(n)} = & \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_n \cdots \int_{t_0}^{t_2} dt_1 \int dx_n dx'_n \cdots \int dx_1 dx'_1 \\
& \times K_0(x, t; x'_n, t_n) \langle x'_n | \hat{V}(t_n) | x_n \rangle K_0(x_n, t_n; x'_{n-1}, t_{n-1}) \langle x'_{n-1} | \hat{V}(t_{n-1}) | x_{n-1} \rangle \\
& \times \cdots \times \langle x'_1 | \hat{V}(t_1) | x_1 \rangle K_0(x_1, t_1; x_0, t_0).
\end{aligned} \tag{48}$$

This is the fully general expression, and it reduces to the local result upon substituting $\langle x'_k | \hat{V}(t_k) | x_k \rangle = V(x_k, t_k) \delta(x'_k - x_k)$ and performing the dx'_k integrations.

Physical interpretation: sum over histories

The Born series has a beautiful physical interpretation. Each n -th order term represents all processes in which the particle undergoes exactly n scatterings. The spatial integrations sum over all possible positions $x_1, \dots, x_n$ at which the scatterings could occur, and the time integrations sum over all possible (time-ordered) instants $t_1, \dots, t_n$ of these events. Between scatterings, the particle propagates freely according to K_0 . The full propagator is thus the sum over all possible “histories” of the particle, weighted by the appropriate free propagation amplitudes and scattering strengths:

$$K = \underbrace{K_0}_{\text{no scattering}} + \underbrace{K_0 V K_0}_{\text{single scattering}} + \underbrace{K_0 V K_0 V K_0}_{\text{double scattering}} + \cdots \tag{49}$$

This is reminiscent of (and indeed closely related to) the Feynman path integral, where the propagator is obtained by summing over all possible paths weighted by $e^{iS/\hbar}$. Here, we are instead summing over all possible scattering sequences, which is the perturbative expansion of the same object. The connection becomes precise in quantum field theory, where each term in the Dyson series corresponds to a set of Feynman diagrams.